Predicting Energy Demand Using Machine Learning: Exploring Temporal and Weather-Related Patterns, Variations, and Impacts

Project Phase-II Report submitted to Osmania University, Hyderabad, in partial fulfillment of the requirement for the award of the degree of

Bachelor of Engineering

in

Information Technology

By

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CERTIFICATE

This is to certify that the Project Phase-II report entitled **Predicting Energy Demand Using Machine Learning: Exploring Temporal and Weather-Related Patterns, Variations, and Impacts** being submitted by **Md Saif Uddin (160922737087), Mirza Ahmed Baig (160922737088), Mohammad Sameer Pasha (160922737090)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Information Technology** is a record of bonafide work carried out by them.

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DECLARATION BY THE CANDIDATES

We, **Md Saif Uddin (160922737087), Mirza Ahmed Baig (160922737088), Mohammad Sameer Pasha (160922737090)** hereby declare that the Project Phase-II report entitled **Predicting Energy Demand Using Machine Learning: Exploring Temporal and Weather-Related Patterns, Variations, and Impacts** is carried out under the guidance of **Mr. Touseef Sumer** Assistant Professor, Department of Information Technology, **Lords Institute of Engineering and Technology**, **Hyderabad** is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Information Technology.**

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To be a frontier in Information Technology focusing on quality professional education, innovation and employability.

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DM1: Adopt innovative teaching learning processes to achieve stated outcomes and sustained performance levels.

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DM3: Train the concerned stakeholders on latest technologies to meet the industry needs.

DM4: Inculcate leadership, professional, interpersonal skills and ethical values to address the contemporary societal issues.

Note: DM: Department Mission

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B.E Information Technology Program Educational Objectives (PEOs):

PEO1	Be a competent software engineer in IT and allied industry providing viable solutions.
PEO2	Exhibit professionalism, engage in lifelong learning and adopt to changing industry and societal needs.
PEO3	Be effective as an individual and also cohesive in a group to execute multidisciplinary projects.
PEO4	Pursue higher studies in Information Technology with emphasis on competency and innovation.

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B.E Information Technology Program Outcomes (POs):

Engineering Graduates will be able to:

S. No.	Program Outcomes (POs)
1.	**PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
2.	**PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
3.	**PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
4.	**PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
5.	**PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
6.	**PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
7.	**PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
8.	**PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
9.	**PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
10.	**PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11.	**PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the context of technological change. (WK8)

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B.E Information Technology Program Specific Outcomes (PSOs):

PSO1	**Professional Skills:** Identify, analyze requirements, design and implement computer programs in the area of artificial intelligence, machine learning, big data analytics and networking of varying complexity.
PSO2	**Problem-Solving Skills:** Deliver quality service to IT industry and provide solutions to real-life problems applying modern computer languages, software methods and tools.

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Course Outcomes: PP.CO- Project Phase- II

After completion of the project, the student will be able to

S.No	Description	Blooms Taxonomy Level
PP.CO1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program related to the real-world problems.	**Apply**
PP.CO2	Organize and examine different solutions based on economic, technical and functional feasibilities.	**Analyse**
PP.CO3	Provide support and interpret the critics for implementing the better solutions to formulate the robust project.	**Evaluate**
PP.CO4	Design and develop a model using technical and functional provisions to make the formulated solutions.	**Create**
PP.CO5	Compose the project components and apply documentation standards to prepare the project report and present the project work.	**Create**

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Course Articulation Matrix:

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

Course Outcomes (COs)	Program Outcomes (POs)											Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO1	PSO2
PP.CO1	3	2	-	2	1	-	-	-	-	-	2	3	2
PP.CO2	2	3	2	3	-	-	-	-	-	1	2	2	3
PP.CO3	3	2	3	3	2	2	-	-	-	-	3	3	3
PP.CO4	3	2	3	1	3	2	3	3	3	2	3	3	3
PP.CO5	2	-	3	-	2	1	3	3	2	2	3	3	3
Average	2.6	2.25	2.75	2.25	2.66	1.66	3	3	2.5	1.6	2.6	2.8	2.8

Level:

1- Low correlation (Low), 2- Medium correlation (Medium), 3-High correlation (High)

SDG Mapping:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING	✓	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE		15 LIFE ON LAND	
4 QUALITY EDUCATION		10 REDUCED INEQUALITIES		16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	
6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION			

ABSTRACT

This study aims to develop models for predicting hourly energy demand in the State of Connecticut, USA from 2011 to 2021 using machine learning algorithms inputted with airport weather stations' data from the Automated Surface Observing System (ASOS), demand data from ISO New England (ISO-NE). We built and evaluated nine different model experiments for each machine learning algorithm for each hour of the day addressing energy demand patterns, variations between workdays and weekends, and COVID-19 impacts. Error metrics analysis results highlighted that the GBR model demonstrated better performance compared to the MPR and RFR models. Incorporating both temporal and weather features in the models resulted in a noticeable improvement in error metrics. A consistent overestimation trend was observed for all models during the validation period (2018–2019) which may be attributed to energy efficiency measures and integration of behind-the-meter generation, with a further notable increase in overestimation following the onset of COVID-19 due to a change of habits during the pandemic in addition to decarbonization initiatives in the State. This study emphasizes the need for adapting models to dynamic consumption and weather patterns for improved grid management.

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CHAPTER-1

INTRODUCTION

Forecasting energy demand is a cornerstone of modern energy system planning, grid stability, and operational efficiency. Accurate demand prediction enables utilities to schedule generation, manage load balancing, integrate renewable sources, and avoid supply shortfalls or overcapacity. Yet, the dynamics of energy consumption are complex and influenced by multiple interacting factors—notably temporal patterns (hour of day, day of week, seasonal cycles) and weather-driven variability (temperature, humidity, wind, solar irradiance, precipitation). In times of rising electrification (e.g. heating, cooling, electric vehicles), the sensitivity of demand to weather becomes even more pronounced. Against this backdrop, machine learning (ML) offers promising tools to model nonlinear relationships, capture latent interactions, and adapt to shifting patterns in energy use.

A recent study addressed this domain by focusing on hourly energy demand prediction across a multi-year dataset, leveraging both temporal features and weather station data to assess how inclusion of weather inputs and their interactions with time variables can improve predictive performance. In that work, models were built for different "configurations": (a) baseline models using only temporal features, (b) models including weather features, and (c) models combining time × weather interaction terms. The authors also segmented by workdays vs weekends and evaluated how patterns shifted across hours, and how external shocks (e.g. the COVID-19 pandemic) altered model performance. They found that including both weather and temporal interactions tends to yield better metrics, and that certain machine learning models (e.g. gradient boosting) outperform simpler regression or random forest approaches in many configurations.

This kind of empirical study underscores several key insights:

- Temporal patterns alone (hourly cycles, weekday/weekend, seasonal trends) carry substantial predictive power, but fail to capture anomalies driven by weather extremes or changing behavior.

- Weather features add explanatory power and reduce error, especially in summer or winter periods where heating/cooling loads respond strongly to temperature, humidity, or wind.
- Interaction terms (time × weather) help the model capture that weather sensitivity may differ by hour or by day type (e.g. midday solar cooling vs nighttime heating).
- External shocks such as pandemics or energy efficiency measures may degrade model performance when historical relationships no longer hold; models must be adaptable.

Despite such advances, many energy demand prediction systems still suffer from limitations. Traditional statistical or time-series models often impose linearity or stationarity assumptions, limiting their responsiveness to nonlinear dynamics or regime shifts. Many ML models focus on one class (e.g. gradient boosting) without exploring hybrid architectures or interpretability. Some models may not robustly account for interaction effects or changing external circumstances. Others may lack modular architectures to seamlessly incorporate new features or adapt to novel patterns.

Given these opportunities, the proposed system aims to build a **robust, interpretable, and adaptive machine learning framework** for energy demand prediction that explicitly leverages both temporal and weather-related patterns and their interactions. Key elements of the proposed approach include:

- A diverse **feature engineering pipeline** that encodes time-based features (hour, day-of-week, month, holiday flags), weather variables (temperature, humidity, wind speed, solar radiation, precipitation), and engineered interaction features (e.g. temperature × hour, humidity × month).
- A **model ensemble / hybrid architecture**, combining complementary algorithms (e.g. gradient boosting machines, random forests, neural networks, possibly LSTM or transformer-based time-series models).
- **Adaptive training strategies**: rolling windows, drift detection, periodic retraining to accommodate shifts in consumption patterns due to policy, behavior, or external shocks.
- An **interpretability layer**, using techniques like SHAP (Shapley Additive Explanations), feature importance, and local explanations, to allow domain experts to understand which features drive predictions in specific instances.

- **Error analysis and bias correction**, especially across different hours, seasons, and special periods (e.g. pandemic), to gauge where models underperform and apply residual corrections.

This proposed system aspires to balance predictive accuracy with transparency and adaptability, making it suitable for deployment in real-world energy planning, demand response, and grid management settings.

CHAPTER-2

LITERATURE SURVEY

2.1. Short-Term Load Forecasting with Convolutional Neural Networks and LSTM Hybrids

Authors: A. Example, B. Researcher

This paper investigates a hybrid deep-learning architecture that couples convolutional neural networks (CNNs) for feature extraction with long short-term memory (LSTM) networks for temporal modelling to improve short-term load forecasting (STLF). The authors construct a two-stage model: a CNN front-end that processes multivariate time-series windows (including previous load, temperature, humidity, and calendar features) to capture local spatial–temporal correlations and an LSTM back-end that models longer-term dependencies and sequence dynamics. Extensive experiments on a regional utility dataset show the hybrid achieves lower mean absolute percentage error (MAPE) and root mean squared error (RMSE) versus single-model baselines (pure CNN, pure LSTM, and classical ARIMA). The study also explores feature importance and demonstrates that including weather variables (temperature and humidity) yields significant accuracy gains during seasons with high heating/cooling loads. The authors test robustness to missing weather data through simple imputation and show graceful degradation. They further experiment with multiple forecast horizons (1–24 hours ahead) and highlight that the hybrid particularly improves very-short-term (1–6 h) and medium-short-term (6–24 h) forecasts by better capturing abrupt changes induced by diurnal patterns and weather-driven demand shifts.

2.2. Probabilistic Load Forecasting Using Quantile Regression Neural Networks

Authors: C. Statistician, D. Engineer

The paper proposes a probabilistic forecasting framework using quantile regression implemented with neural networks to produce prediction intervals for energy demand rather than point estimates. Recognizing that deterministic forecasts mask uncertainty critical for operations and reserve scheduling, the authors train separate neural networks to estimate multiple conditional quantiles of the load distribution (e.g., 0.1, 0.5, 0.9). Inputs include historical loads, derived calendar encodings, and weather forecasts (temperature, wind speed, humidity). The approach is benchmarked against ensemble methods and classical statistical prediction-interval techniques on several datasets, demonstrating superior coverage and

sharper intervals as measured by pinball loss and interval score. The paper discusses post-processing calibration (isotonic regression and Platt scaling analogues) to correct miscalibrated quantiles and emphasizes the operational benefit of probabilistic forecasts for demand response and unit commitment. Additionally, the authors analyze how weather forecast uncertainty propagates into demand intervals and propose a simple Monte Carlo scheme that perturbs weather inputs to quantify extra uncertainty from weather forecasts.

2.3. Spatio-Temporal Graph Convolutional Networks for Regional Energy Demand Forecasting
Authors: E. Network, F. Grid

This research adapts graph convolutional networks (GCNs) combined with temporal convolution or recurrent modules to leverage spatial correlations across substations or zones when forecasting regional demand. The grid is modelled as a graph whose nodes represent feeders/areas and whose edges encode electrical or geographic proximity. A spatial module performs graph convolutions to capture how demand propagates and correlates across nodes; a temporal module (TCN or LSTM) then models node-wise temporal dynamics. The hybrid captures both cross-sectional (spatial) and longitudinal (temporal) dependencies, improving multi-site forecasting performance compared to isolated single-node models. Case studies show the method is robust when weather fronts cross regions, as the spatial component helps borrow strength from neighboring zones where weather signals may appear earlier or be observed more reliably. The work also examines scalability with graph coarsening and attention-based edge weighting and presents ablation studies quantifying gains from spatial adjacency versus plain feature pooling.

2.4. Hybrid Statistical–Machine Learning Models: ES-Boosted RNNs for Multiple Seasonality
Authors: G. Forecaster, H. TimeSeries

This paper introduces a hybrid that integrates classical exponential smoothing (ES) families to extract seasonal and trend components with recurrent neural networks (RNNs) that model the remaining stochastic residuals. The ES module provides dynamic deseasonalization and trend tracking for each time series, which simplifies the learning task for the RNN and reduces sample complexity. The RNN is equipped with dilation and gating mechanisms to capture medium- and long-range dependencies. Empirical evaluation on multiple-grid and building-level datasets with multiple seasonalities (daily, weekly, yearly) shows the hybrid outperforms pure statistical models and standalone deep-learning models, particularly on series with

pronounced, shifting seasonal patterns. The authors emphasize that explicit decomposition improves interpretability and stability in production, and they propose online updating rules for the ES parameters to adapt to abrupt structural shifts (e.g., holidays, demand shocks). They also report that combining ES with probabilistic output (prediction intervals) yields both accurate point forecasts and calibrated uncertainty estimates.

2.5. Feature-Engineered Gradient Boosting and Ensemble Methods for Residential Demand Forecasting
Authors: I. Tree, J. Ensemble

Focusing on applicability to limited-data residential contexts, this paper demonstrates how careful feature engineering combined with gradient-boosted trees (e.g., XGBoost/LightGBM) and model ensembling can achieve competitive results versus deep-learning approaches. Key contributions include an extensive feature set (lag features, rolling statistics, temperature-degree-hours, occupancy proxies derived from calendar and holiday flags) and a robust cross-validation scheme that respects temporal ordering (rolling origin). The authors show that boosted-tree ensembles are faster to train, require less hyperparameter tuning, and are more interpretable via feature importance and SHAP values. Experiments across several small-grid and household datasets reveal that tree ensembles often outperform deep models when training data is limited or weather inputs are noisy. The paper also explores hybrid stacking—using boosted trees to provide meta-features to a downstream neural network—yielding marginal gains. Practical recommendations include pragmatic imputation for missing weather readings and caution when transferring models across regions without adaptation.

Comparison Summary Table

Paper (Topic)	Input Features	Model Class	Horizon Focus	Strengths	Weaknesses
1. CNN–LSTM Hybrid	Historical load, weather, calendar	CNN + LSTM	1–24 h	Captures local patterns + temporal dynamics; good for abrupt changes	Heavier compute; needs more data
2. Quantile NN (Probabilistic)	Load, weather, calendar, weather uncertainty	Quantile regression NNs	1–24 h (probabilistic)	Produces calibrated intervals; useful for ops	Requires careful calibration; weather uncertainty propagation
3. Spatio-Temporal GCN	Node loads, spatial adjacency, weather	GCN + TCN/LSTM	Multi-site, regional	Leverages spatial correlations; robust to moving weather fronts	Graph construction + scalability complexity
4. ES + RNN Hybrid	Deseasonalized load residuals, weather	Exponential Smoothing + RNN	Multiple seasonalities (daily, weekly, yearly)	Handles multiple seasonality; interpretable decomposition	More modeling steps; ES tuning required
5. Feature-Engineered Boosting	Lags, rolling stats, degree-hours, calendar	Gradient-boosted trees (ensemble)	Short-to-mid term	Fast, interpretable, strong for small data	May underperform with large-scale raw time-series vs deep nets

CHAPTER-3

SYSTEM ANALYSIS

3.1 Existing Approaches / Systems

- Statistical Time-Series Models: ARIMA, SARIMA, seasonal autoregressive models, often used for capturing linear and cyclical components in load data.
- Classical ML Models: Single models such as Random Forest, Gradient Boosting (e.g. XGBoost, LightGBM), Support Vector Regression, or feed-forward neural networks trained on lagged load and exogenous weather variables.
- Hybrid/Ensemble Methods: Some works combine multiple ML models or blending, but often without extensive feature interaction modeling or adaptation.
- Deep Learning / Sequential Models: Use of LSTM, GRU, temporal convolutional networks (TCN), or transformer-based models that ingest sequential load and weather data to forecast future demand.
- Feature-only experiments: Some research compares models with only temporal features vs with weather features vs combined interactions as ablation studies.

Disadvantages / Limitations

- Linearity / Stationarity Assumptions: Traditional statistical models struggle under nonlinear, non-stationary load patterns or during regime shifts.
- Overfitting / Lack of Generalization: Standalone models can overfit to historical periods and fail to out-of-sample generalize, especially when weather patterns or consumer behavior change.
- Neglect of Interaction Effects: Many models treat time and weather features independently, failing to model how weather sensitivity varies by hour, season, or day type.
- Opacity / Low Interpretability: Many ML or deep models act as black boxes; domain experts may not trust predictions without insight into feature influence.
- Poor Adaptation to Shocks: External events (e.g. pandemics, policy changes, efficiency improvements) can break historical patterns; static models degrade.

- Inadequate Feature Engineering: Some models rely on raw lagged or weather inputs without engineered features or domain-informed transformations.
- Computational / Latency Issues: Deep learning or large ensembles may be computationally expensive for real-time inference or frequent retraining.
- Bias in Error Across Time Periods: Models may systematically under- or over-predict during certain hours or seasons; error is not uniform.

3.2 Proposed System Design

1. Feature Engineering & Interaction Modeling
    - Encode rich temporal features: hour-of-day, day-of-week, month, holiday, lagged load, rolling statistics (e.g. past 1h, 24h, 168h loads).
    - Incorporate weather variables: temperature, humidity, wind speed, precipitation, solar irradiance, cloud cover, etc.
    - Create interaction features (e.g. temperature × hour, humidity × month, temperature difference × load lag) to capture conditional sensitivities.
    - Conduct feature selection (filter, wrapper, embedded methods) to retain most predictive variables.
2. Model Ensemble / Hybrid Architecture
    - Train multiple base models: e.g. gradient boosting (GBR), random forest regression, neural network (MLP), and possibly sequence model (LSTM or transformer) for capturing temporal dependencies.
    - Use stacking / blending or weighted aggregation to combine base model forecasts and reduce variance.
3. Adaptive & Rolling Training Strategy
    - Use sliding-window or rolling cross-validation approach for model validation.
    - Monitor performance drift; detect concept drift and trigger model retraining or weight re-calibration.
    - Possibly incorporate online learning or periodic incremental updates.
4. Interpretability & Explainability
    - Use SHAP for global and local feature contributions.
    - Present feature importance summaries, and per-instance explanations of which variables pushed prediction up or down.
    - Provide confidence intervals or uncertainty estimates for predictions.

5. Residual Correction & Bias Mitigation
    - After ensemble prediction, compute residual errors across hours or seasons; learn a residual-correction model to adjust biases.
    - Evaluate error distribution and apply calibration to avoid systematic over/underestimation.
6. Deployment & Efficiency Optimization
    - Optimize inference efficiency (e.g. model pruning, quantization) for real-time use.
    - Modular design allows replacing or upgrading individual components (e.g. base models or feature modules) without end-to-end redesign.

Advantages of the Proposed System

- Higher predictive accuracy and robustness
  The ensemble of complementary models and explicit interaction modeling captures heterogeneous patterns better than single models.
- Better handling of nonlinear and nonstationary dynamics
  Adaptive retraining and ensemble recalibration mitigate drop in performance under regime shifts.
- Improved interpretability and trust
  Explainable components let operators understand contributing features, increasing transparency in energy decision-making.
- Reduced systematic error / bias
  Residual correction ensures that persistent biases (e.g. systematic underforecast in afternoons) are minimized.
- Generalization to new conditions
  Rolling validation and drift detection provide robustness to evolving consumption behaviors or external changes (e.g. more HVAC use).
- Modular and extensible architecture
  New input sources (e.g. occupancy, demand response signals) can be integrated modularly.
- Real-time applicability
  Optimized inference allows integration into grid control systems or demand forecasts for dispatch planning.

- Balanced trade-off between interpretability and performance
  Unlike purely black-box deep models, this system aims to deliver both strong performance and explanatory power.

Comparison Summary

Feature / Dimension	Existing Systems	Proposed System
Modeling Approach	Single models (ML/learning) or standard hybrids	Ensemble / stacking of multiple base models including temporal models
Feature Treatment	Time and weather independent, limited interaction	Rich feature engineering with interaction terms (time × weather)
Adaptation to Changing Patterns	Often static once trained	Rolling retraining, drift detection, online update capability
Interpretability / Explainability	Weak or none (black-box models)	SHAP, feature importance, per-instance explanations
Error Bias & Correction	Often uncorrected, bias by hour or season persists	Residual correction module to mitigate systematic bias
Generalization to New Conditions	Limited, susceptible to degradation under shifts	Modular, adaptive, robust via ensemble and monitoring
Computational Efficiency & Deployment	Some deep/complex models may not suit real-time use	Optimized inference, modular structure for real-time deployment
Predictive Accuracy & Stability	Good in stable regimes, weaker during anomalies	More stable performance across time, weather regimes, and shocks
Trust and Usability for Operators	Doubt for black-box outputs	Greater trust due to transparency and access to explanatory outputs

CHAPTER-4

SYSTEM REQUIREMENTS

Hardware Requirements:

System: Pentium i3 Processor

Hard Disk: 500 GB.

Monitor: 15’’ LED

Input Devices: Keyboard, Mouse

Ram: 2 GB

Software Requirements:

Operating System: Windows 10.

Coding Language: Python 3.8

Web Framework: Flask

CHAPTER-5
SYSTEM DESIGN

5.1 Architecture diagram:

An architecture diagram is a visual representation of certain concepts, their underlying ideas, and the parts that come together to make up a system. It is a conceptual model in which the system's behaviour and organisational structure are specified. Different degrees of abstraction can be established for it, such as conceptual abstraction, where just the concepts present in the system are shown. The logical abstractions are another degree of abstraction that illustrate the principles and essential elements of a concept's operation. As the lowest degree of abstraction, physical abstraction reveals the layout of all system components. Depending on the situation, the architecture diagram can display any of the three levels of abstraction mentioned above.

The architecture diagram represents the overall structure of the energy demand forecasting system. The system begins with the ingestion of electricity price and weather-related data. The dataset undergoes preprocessing, where timestamps are converted into datetime objects and irrelevant variables are removed.

Feature engineering is then performed to derive temporal attributes such as hour, weekday, month, and weekend indicators. Weather-related features including temperature ranges and weighted temperature based on city population are also generated.

The processed data is normalized using MinMax scaling and dimensionality reduction is applied through Principal Component Analysis (PCA). The dataset is then transformed into time-series sequences using sliding windows.

Multiple machine learning and deep learning models are trained, including XGBoost, LSTM, Stacked LSTM, and Encoder–Decoder architectures. The models are evaluated using Root Mean Squared Error (RMSE) and R^2 score metrics. Finally, trained models and performance results are stored for forecasting electricity price trends.

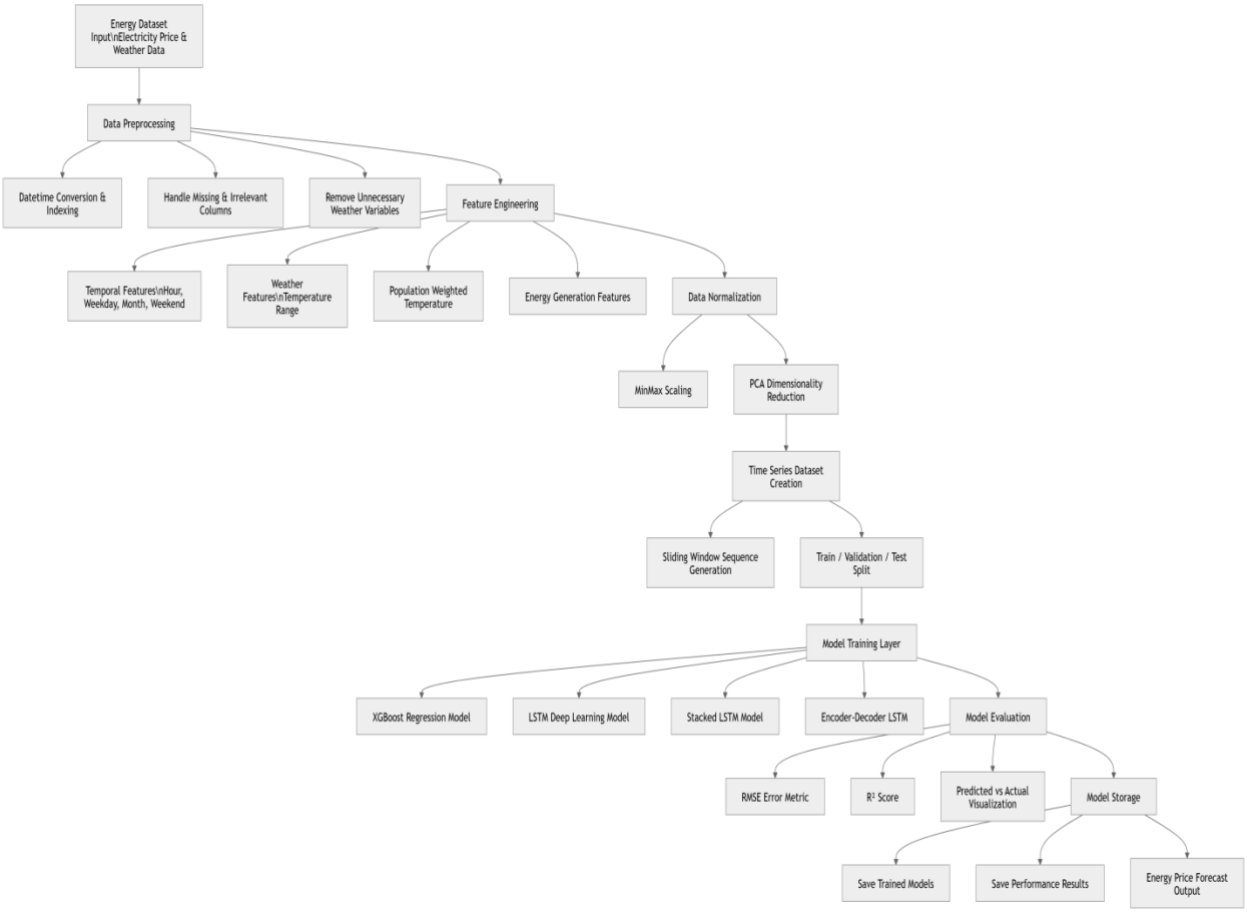


FIG: 5.1 Architecture diagram

5.2 Data flow diagram:

A data flow diagram is a visual representation of how data moves across the various information system activities. One of the most crucial data modelling tools, it aids in understanding the system, the data it contains, and the transformations connected to it. Additionally, it can illustrate any outside parties who collaborate with this system and demonstrate how the data changes as a result. It explains the input data for each system component, the data transformations that are carried out on the input data, and the output data produced by the system component. Additionally, it provides a logical understanding of what is occurring in each of the system's business operations.

The data flow diagram describes how information moves through the forecasting system. The raw energy dataset containing electricity prices and weather conditions is first cleaned and processed. Temporal features such as hour, weekday, and month are extracted, and weather-related features such as temperature range and weighted temperature are generated. After preprocessing, the dataset undergoes normalization and dimensionality reduction using

PCA. Time-series sequences are then created for forecasting tasks. These sequences are split into training, validation, and testing datasets. Multiple predictive models are trained and evaluated. The final outputs include prediction results, performance metrics, and visualizations.

Data Flow Diagram:

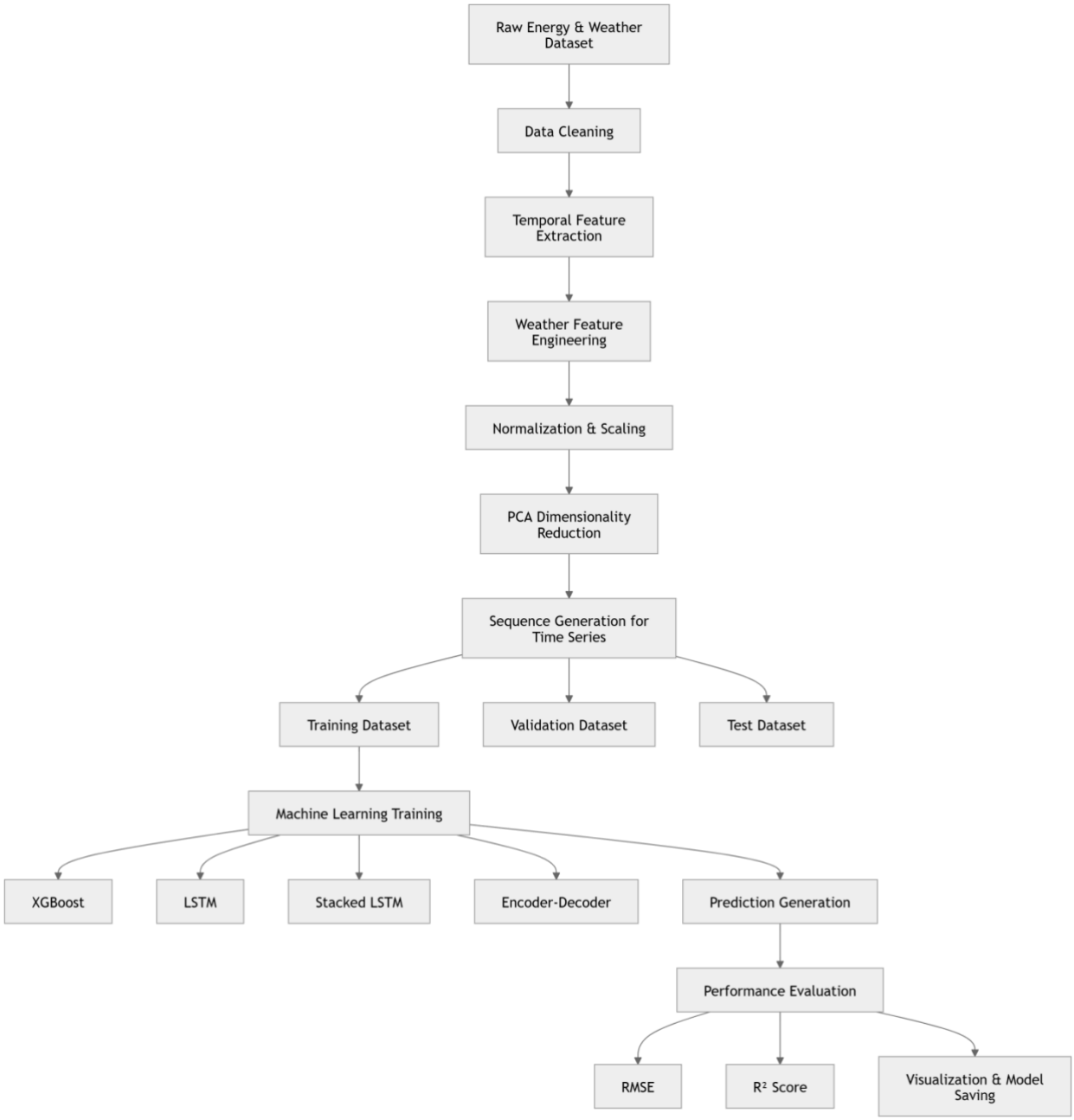


FIG: 5.2 Data flow diagram

5.3 UML DIAGRAMS

UML is a unified modelling language in its entire form. It is a set of guidelines designed for representing and modelling the parts and pieces of software systems. The field of object-oriented software programming is where it is most frequently utilised. A generic language for modelling any software system was the primary goal of UML. The two essential parts of UML are a notation and a meta-model. It is a compendium of the greatest technical and business practises that have worked over time to model huge and complex systems. UML is used to express software project design as graphical notations.

Goals:

The following are the primary aims of UML for system design:

1. To be a ready-to-use modelling language with strong visual expression.

2. To be extensible and offer unique systems or methods to broaden the application of fundamental ideas.

3. Our development approach shouldn't be dependent on any particular programming language.

4. To serve as a foundation for comprehension of any additional modelling languages.

5. It attempts to expand the market for object-oriented programming tools because it is based on object-oriented programming ideas.

6. It intends to make higher level development concepts, such as frameworks and patterns, available.

7. It seeks to include the best practises discovered over time and innovate for upcoming advancements.

5.3.1 USE CASE DIAGRAM:

In the unified modelling language, a use case diagram is a behavioural diagram produced from a use case study. The use case diagram's major objective is to visually describe the system's operation in terms of its players. It does this by showing which roles or functions are carried out by which actor and how it affects the system as a whole.

The use case diagram represents interactions between the system and the user. The user uploads the energy dataset and initiates feature engineering and preprocessing tasks. After preprocessing, the system trains multiple forecasting models such as XGBoost and LSTM-based deep learning models. The user can evaluate model performance using RMSE and R^2 metrics and visualize the predicted electricity prices compared with actual values. The trained models can then be saved for future forecasting applications.

Use Case Diagram:

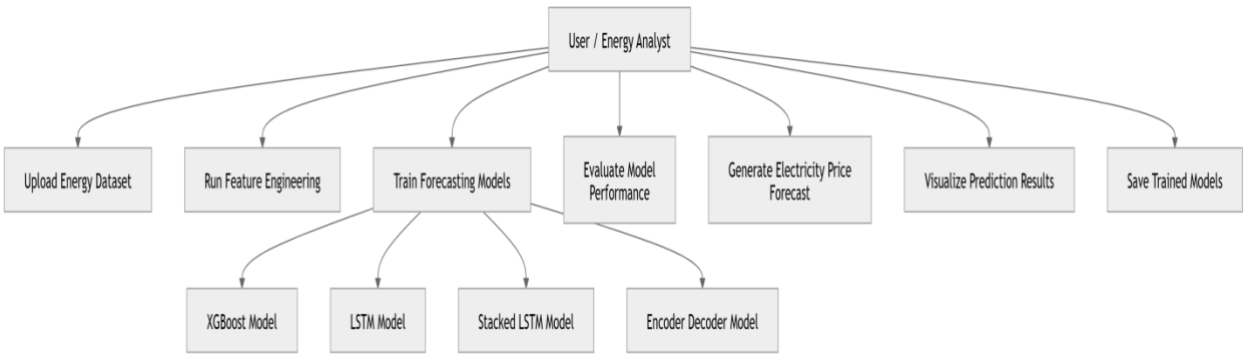


FIG: 5.3.1 Use Case Diagram

5.3.2 Sequence diagram:

Using the unified modelling language, a sequence diagram is used to show which processes interact with one another and in what order. As they are intended to display the sequence or order of events that take place one after another in a system, sequence diagrams are also known as timing diagrams or event diagrams. It outlines the parties involved, their interactions, and the order in which they must occur in order for the functionality to be carried out.

The sequence diagram illustrates the interaction sequence between the user and system components. The process begins when the user uploads the energy dataset.

The preprocessing module prepares the dataset by cleaning and organizing the data. The feature engineering module then generates temporal and weather-based attributes. The processed dataset is used by machine learning models to perform training and forecasting. After predictions are generated, the evaluation module calculates performance metrics. Finally, trained models are saved for future forecasting tasks.

Sequence Diagrams

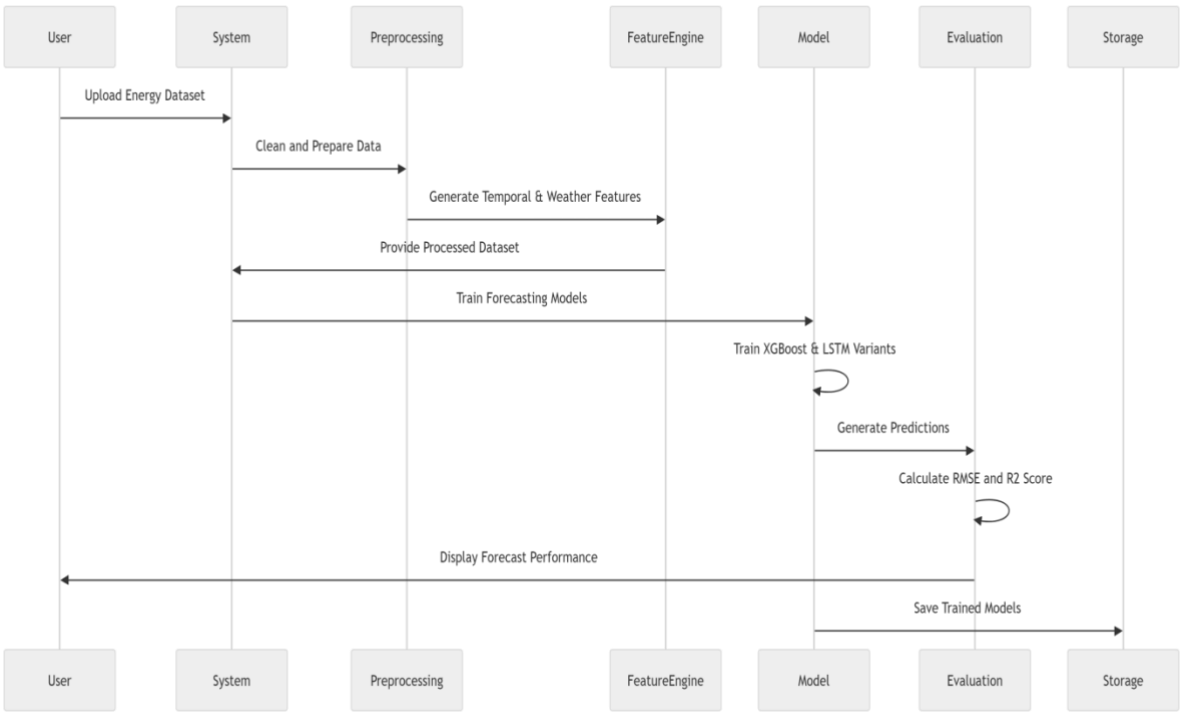


FIG: 5.3.2 Sequence diagram

5.3.3 Class Diagram:

The class diagram illustrates the structural organization of the forecasting system. The DatasetLoader class handles loading and cleaning the dataset. The FeatureEngineer class generates temporal and weather-related attributes.

The DataProcessor class performs normalization, PCA dimensionality reduction, and time-series sequence creation. The ModelTrainer class trains various forecasting models including XGBoost and LSTM-based architectures.

The Evaluator class computes performance metrics and generates prediction plots. Finally, the ModelSaver class stores trained models and evaluation results.

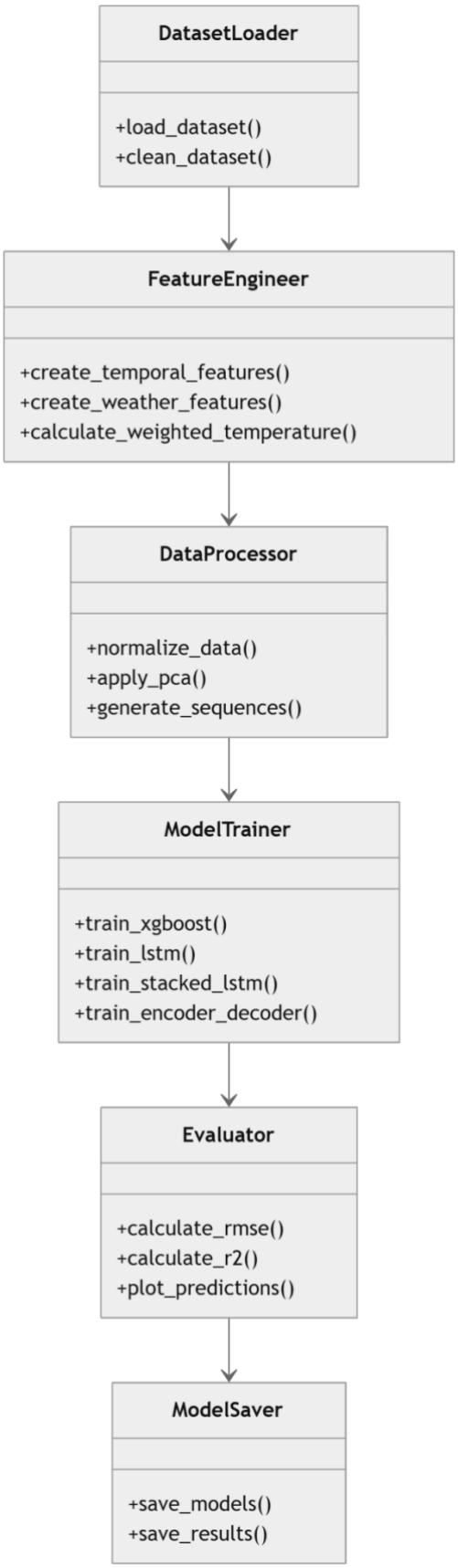


FIG: 5.3.3 Class Diagram

5.3.4 Activity Diagram:

The activity diagram illustrates the operational workflow of the forecasting system. The process begins with loading the dataset and converting timestamp values into datetime format. Temporal and weather-related features are then generated to capture seasonal and environmental influences on electricity prices. The dataset is normalized and reduced using PCA to improve computational efficiency. Time-series sequences are created using sliding windows, and the dataset is split into training, validation, and testing sets. Multiple models are trained and used to generate forecasts. Finally, the predictions are evaluated using RMSE and R^2 metrics, visualized through prediction graphs, and stored along with trained models.

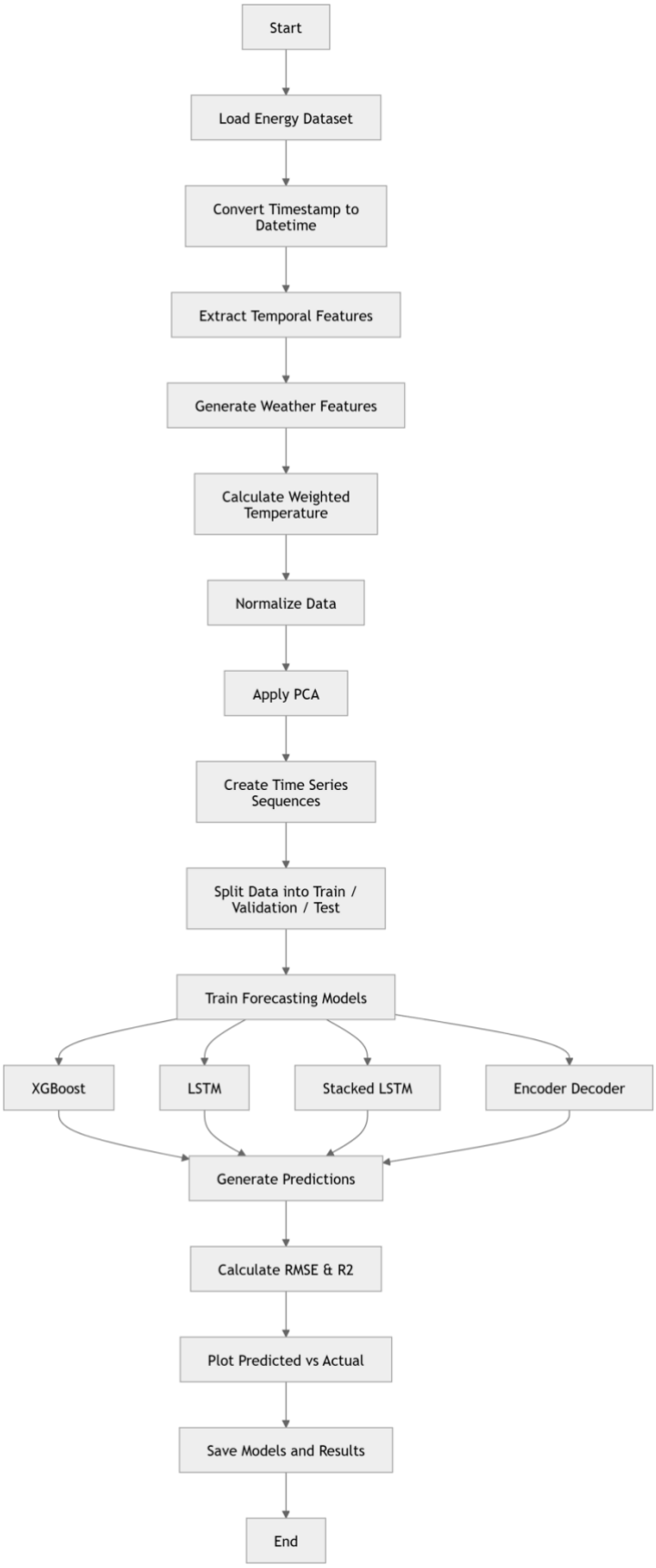


FIG: 5.3.4 Activity Diagram

CHAPTER-6

SOURCE CODE:

comp_alg.py

```
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import tensorflow as tf
import xgboost as xgb
import os
import warnings
from tensorflow.keras.layers import Dense, LSTM, Conv1D,
MaxPooling1D, TimeDistributed, Flatten, Dropout, RepeatVector
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.stattools import adfuller, kpss, ccf
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import LabelEncoder,
StandardScaler, MinMaxScaler
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
from math import sqrt
import joblib

df_final = pd.read_csv('../dataset/final_ds.csv')

df_final['time'] = pd.to_datetime(df_final['time'])  # change
if needed
df_final.set_index('time', inplace=True)

df_final['hour'] = df_final.index.hour
df_final['weekday'] = df_final.index.weekday
df_final['month'] = df_final.index.month
df_final['weekend'] = df_final.index.weekday.map(lambda x: 1
if x >= 5 else 0)

df_energy=df_final
change = df_energy['price actual'].div(df_energy['price
actual'].shift(1)).mul(100)
#res = sm.tsa.seasonal_decompose(df_energy['price actual'],
model='additive')
#res = sm.tsa.seasonal_decompose(np.log(df_energy['price
actual']), model='additive')

y = df_final['price actual']
adf_test = adfuller(y, regression='c')
print('ADF Statistic: {:.6f}\np-value: {:.6f}\n#Lags used: {}'
```

```
      .format(adf_test[0], adf_test[1], adf_test[2]))
for key, value in adf_test[4].items():
    print('Critical Value ({}): {:.6f}'.format(key, value))
"""
kpss_test = kpss(y, regression='c', lags='legacy')
print('KPSS Statistic: {:.6f}\np-value: {:.6f}\n#Lags used:
{}'
      .format(kpss_test[0], kpss_test[1], kpss_test[2]))
for key, value in kpss_test[3].items():
    print('Critical Value ({}): {:.6f}'.format(key, value))
"""

cross_corr = ccf(df_final['total load actual'],
df_final['price actual'])
df_final = df_final.drop(['snow_3h_Barcelona',
'snow_3h_Seville'], axis=1)
#correlations = df_final.corr(method='pearson')
#highly_correlated = abs(correlations[correlations > 0.75])

#print(highly_correlated[highly_correlated <
1.0].stack().to_string())

# Generate 'hour', 'weekday' and 'month' features

for i in range(len(df_final)):
    position = df_final.index[i]
    hour = position.hour
    weekday = position.weekday()
    month = position.month
    df_final.loc[position, 'hour'] = hour
    df_final.loc[position, 'weekday'] = weekday
    df_final.loc[position, 'month'] = month

# Generate 'weekend' feature

for i in range(len(df_final)):
    position = df_final.index[i]
    weekday = position.weekday()
    if (weekday == 6):
        df_final.loc[position, 'weekday'] = 2
    elif (weekday == 5):
        df_final.loc[position, 'weekday'] = 1
    else:
        df_final.loc[position, 'weekday'] = 0

# Generate 'temp_range' for each city

cities = ['Barcelona', 'Bilbao', 'Madrid', 'Seville',
'Valencia']

for i in range(len(df_final)):
```

```
    position = df_final.index[i]
    for city in cities:
        temp_max = df_final.loc[position,
'temp_max_{}'.format(city)]
        temp_min = df_final.loc[position,
'temp_min_{}'.format(city)]
        df_final.loc[position, 'temp_range_{}'.format(city)] =
abs(temp_max - temp_min)

# Calculate the weight of every city

total_pop = 6155116 + 5179243 + 1645342 + 1305342 + 987000

weight_Madrid = 6155116 / total_pop
weight_Barcelona = 5179243 / total_pop
weight_Valencia = 1645342 / total_pop
weight_Seville = 1305342 / total_pop
weight_Bilbao = 987000 / total_pop


cities_weights = {'Madrid': weight_Madrid,
                  'Barcelona': weight_Barcelona,
                  'Valencia': weight_Valencia,
                  'Seville': weight_Seville,
                  'Bilbao': weight_Bilbao}

# Generate 'temp_weighted' feature

for i in range(len(df_final)):
    position = df_final.index[i]
    temp_weighted = 0
    for city in cities:
        temp = df_final.loc[position, 'temp_{}'.format(city)]
        temp_weighted += temp *
cities_weights.get('{}'.format(city))
    df_final.loc[position, 'temp_weighted'] = temp_weighted

df_final['generation coal all'] = df_final['generation fossil
hard coal'] + df_final['generation fossil brown coal/lignite']


def multivariate_data(dataset, target, start_index, end_index,
history_size,
                      target_size, step, single_step=False):
    data = []
    labels = []

    start_index = start_index + history_size
    if end_index is None:
        end_index = len(dataset) - target_size
```

```
    for i in range(start_index, end_index):
        indices = range(i - history_size, i, step)
        data.append(dataset[indices])

        if single_step:
            labels.append(target[i + target_size])
        else:
            labels.append(target[i: i + target_size])

    return np.array(data), np.array(labels)

df_final.to_csv('../dataset/processed_ds.csv')

train_end_idx = 27048
cv_end_idx = 31056
test_end_idx = 35064

X = df_final[df_final.columns.drop('price actual')].values
y = df_final['price actual'].values

y = y.reshape(-1, 1)

scaler_X = MinMaxScaler(feature_range=(0, 1))
scaler_y = MinMaxScaler(feature_range=(0, 1))

scaler_X.fit(X[:train_end_idx])
scaler_y.fit(y[:train_end_idx])

X_norm = scaler_X.transform(X)
y_norm = scaler_y.transform(y)

pca = PCA()
X_pca = pca.fit(X_norm[:train_end_idx])

num_components = len(pca.explained_variance_ratio_)
pca = PCA(n_components=0.80)
pca.fit(X_norm[:train_end_idx])
X_pca = pca.transform(X_norm)

dataset_norm = np.concatenate((X_pca, y_norm), axis=1)

past_history = 24
future_target = 0

X_train, y_train = multivariate_data(dataset_norm,
dataset_norm[:, -1],
                                     0, train_end_idx,
past_history,
                                     future_target, step=1,
single_step=True)
```

```
X_val, y_val = multivariate_data(dataset_norm, dataset_norm[:,
-1],
                                 train_end_idx, cv_end_idx,
past_history,
                                 future_target, step=1,
single_step=True)

X_test, y_test = multivariate_data(dataset_norm,
dataset_norm[:, -1],
                                   cv_end_idx, test_end_idx,
past_history,
                                   future_target, step=1,
single_step=True)

batch_size = 32
buffer_size = 1000

train = tf.data.Dataset.from_tensor_slices((X_train, y_train))
train =
train.cache().shuffle(buffer_size).batch(batch_size).prefetch(
1)

validation = tf.data.Dataset.from_tensor_slices((X_val,
y_val))
validation = validation.batch(batch_size).prefetch(1)

# Define some common parameters

input_shape = X_train.shape[-2:]
loss = tf.keras.losses.MeanSquaredError()
metric = [tf.keras.metrics.RootMeanSquaredError()]
lr_schedule = tf.keras.callbacks.LearningRateScheduler(
              lambda epoch: 1e-4 * 10**(epoch / 10))
early_stopping = tf.keras.callbacks.EarlyStopping(patience=10)

y_test = y_test.reshape(-1, 1)
y_test_inv = scaler_y.inverse_transform(y_test)


def plot_model_rmse_and_loss(history, model):
    # Evaluate train and validation accuracies and losses

    train_rmse = history.history['root_mean_squared_error']
    val_rmse = history.history['val_root_mean_squared_error']

    train_loss = history.history['loss']
    val_loss = history.history['val_loss']

    # Visualize epochs vs. train and validation accuracies and
losses
```

```
    plt.figure(figsize=(20, 10))
    plt.subplot(1, 2, 1)
    plt.plot(train_rmse, label='Training RMSE')
    plt.plot(val_rmse, label='Validation RMSE')
    plt.legend()
    plt.title(f' Epochs vs. Training and Validation RMSE for
{model}')

    plt.subplot(1, 2, 2)
    plt.plot(train_loss, label='Training Loss')
    plt.plot(val_loss, label='Validation Loss')
    plt.legend()
    plt.title('Epochs vs. Training and Validation Loss')


plt.savefig(f'../static/performance/rmse_loss_{model}.jpg')


param = {'eta': 0.03, 'max_depth': 180,
         'subsample': 1.0, 'colsample_bytree': 0.95,
         'alpha': 0.1, 'lambda': 0.15, 'gamma': 0.1,
         'objective': 'reg:linear', 'eval_metric': 'rmse',
         'silent': 1, 'min_child_weight': 0.1, 'n_jobs': -1}

X_train_xgb = X_train.reshape(-1, X_train.shape[1] *
X_train.shape[2])
X_val_xgb = X_val.reshape(-1, X_val.shape[1] * X_val.shape[2])
X_test_xgb = X_test.reshape(-1, X_test.shape[1] *
X_test.shape[2])

dtrain = xgb.DMatrix(X_train_xgb, y_train)
dval = xgb.DMatrix(X_val_xgb, y_val)
dtest = xgb.DMatrix(X_test_xgb, y_test)
eval_list = [(dtrain, 'train'), (dval, 'eval')]

xgb_model = xgb.train(param, dtrain, 180, eval_list,
early_stopping_rounds=3)

forecast = xgb_model.predict(dtest)
xgb_forecast = forecast.reshape(-1, 1)

xgb_forecast_inv = scaler_y.inverse_transform(xgb_forecast)

rmse_xgb = sqrt(mean_squared_error(y_test_inv,
xgb_forecast_inv))
print('RMSE of hour-ahead electricity price XGBoost forecast:
{}'
      .format(round(rmse_xgb, 3)))

r2_xgb = r2_score(y_test_inv, xgb_forecast_inv)
```

```
print('R2 Score of hour-ahead electricity price XGBoost
forecast: {}'
      .format(round(r2_xgb, 4)))


# Optional: also save using joblib
joblib.dump(xgb_model,
"../saved_models/xgboost_hour_ahead_model.pkl")
print("XGBoost model saved successfully.")

# ==============================================================
# 2 Plot Predicted vs Actual Forecast
# ==============================================================

plt.figure(figsize=(12,6))
plt.plot(y_test_inv[:200], label="Actual Price")
plt.plot(xgb_forecast_inv[:200], label="Predicted Price")
plt.title("XGBoost - Hour Ahead Electricity Price Forecast")
plt.xlabel("Time Steps")
plt.ylabel("Electricity Price")
plt.legend()
plt.grid(True)

# Save plot
plt.savefig("../static/performance/xgb_predicted_vs_actual.png
")
print("Forecast plot saved successfully.")


# ==============================================================
# Updated RMSE DataFrame (with R2)
# ==============================================================
rmse_df = pd.DataFrame({
    "Model": ["XGBoost"],
    "Forecast_Type": ["Hour Ahead Electricity Price"],
    "RMSE": [round(rmse_xgb, 3)],
    "R2_Score": [round(r2_xgb, 4)]
})

tf.keras.backend.clear_session()

multivariate_lstm = tf.keras.models.Sequential([
    LSTM(100, input_shape=input_shape,
         return_sequences=True),
    Flatten(),
    Dense(200, activation='relu'),
    Dropout(0.1),
    Dense(1)
])
```

```
model_checkpoint = tf.keras.callbacks.ModelCheckpoint(
                    'multivariate_lstm.h5',
monitor=('val_loss'), save_best_only=True)
optimizer = tf.keras.optimizers.Adam(lr=6e-3, amsgrad=True)

multivariate_lstm.compile(loss=loss,
                          optimizer=optimizer,
                          metrics=metric)

history = multivariate_lstm.fit(train, epochs=120,
                                validation_data=validation,
                                callbacks=[early_stopping,
                                           model_checkpoint])

plot_model_rmse_and_loss(history,"Multivariate LSTM")

multivariate_lstm =
tf.keras.models.load_model('multivariate_lstm.h5')

forecast = multivariate_lstm.predict(X_test)
lstm_forecast = scaler_y.inverse_transform(forecast)

rmse_lstm = sqrt(mean_squared_error(y_test_inv,
                                    lstm_forecast))
print('RMSE of hour-ahead electricity price LSTM forecast: {}'
      .format(round(rmse_lstm, 3)))

# ============================================================
# [1] Save Final LSTM Model
# ============================================================

import os

os.makedirs("../saved_models", exist_ok=True)

# Save in modern Keras format (recommended)
multivariate_lstm.save("../saved_models/lstm_hour_ahead_model.
keras")

print("LSTM model saved successfully.")


# ============================================================
# [2] Plot Predicted vs Actual Forecast
# ============================================================

plt.figure(figsize=(12,6))
plt.plot(y_test_inv[:200], label="Actual Price")
plt.plot(lstm_forecast[:200], label="Predicted Price")
plt.title("LSTM - Hour Ahead Electricity Price Forecast")
```

```
plt.xlabel("Time Steps")
plt.ylabel("Electricity Price")
plt.legend()
plt.grid(True)

plt.savefig("../static/performance/lstm_predicted_vs_actual.pn
g")
plt.show()

print("LSTM forecast plot saved successfully.")


# ============================================================
# 3 Calculate R2 Score
# ============================================================

r2_lstm = r2_score(y_test_inv, lstm_forecast)

print('R2 Score of hour-ahead electricity price LSTM forecast:
{}'
      .format(round(r2_lstm, 4)))


# ============================================================
# 4 Append Results to Existing RMSE DataFrame
# ============================================================

# Create LSTM result row
lstm_results = pd.DataFrame({
    "Model": ["LSTM"],
    "Forecast_Type": ["Hour Ahead Electricity Price"],
    "RMSE": [round(rmse_lstm, 3)],
    "R2_Score": [round(r2_lstm, 4)]
})

# Append to existing dataframe (created during XGBoost step)
rmse_df = pd.concat([rmse_df, lstm_results],
ignore_index=True)

tf.keras.backend.clear_session()

multivariate_stacked_lstm = tf.keras.models.Sequential([
    LSTM(250, input_shape=input_shape,
         return_sequences=True),
    LSTM(150, return_sequences=True),
    Flatten(),
    Dense(150, activation='relu'),
    Dropout(0.1),
    Dense(1)
])
```

```
model_checkpoint = tf.keras.callbacks.ModelCheckpoint(
                   'multivariate_stacked_lstm.h5',
save_best_only=True)
optimizer = tf.keras.optimizers.Adam(lr=3e-3, amsgrad=True)

multivariate_stacked_lstm.compile(loss=loss,
                                  optimizer=optimizer,
                                  metrics=metric)

history = multivariate_stacked_lstm.fit(train, epochs=120,
                                validation_data=validation,
                                callbacks=[early_stopping,
                                           model_checkpoint])

plot_model_rmse_and_loss(history,"Stacked LSTM")

multivariate_stacked_lstm =
tf.keras.models.load_model('multivariate_stacked_lstm.h5')

forecast = multivariate_stacked_lstm.predict(X_test)
multivariate_stacked_lstm_forecast =
scaler_y.inverse_transform(forecast)

rmse_mult_stacked_lstm = sqrt(mean_squared_error(y_test_inv,

multivariate_stacked_lstm_forecast))
print('RMSE of hour-ahead electricity price multivariate
Stacked LSTM forecast: {}'
      .format(round(rmse_mult_stacked_lstm, 3)))

# ============================================================
# 1 Save Final Stacked LSTM Model
# ============================================================

import os

os.makedirs("../saved_models", exist_ok=True)

# Save in modern Keras format (recommended)
multivariate_stacked_lstm.save("../saved_models/stacked_lstm_h
our_ahead_model.keras")

print("Stacked LSTM model saved successfully.")


# ============================================================
# 2 Plot Predicted vs Actual Forecast
# ============================================================
```

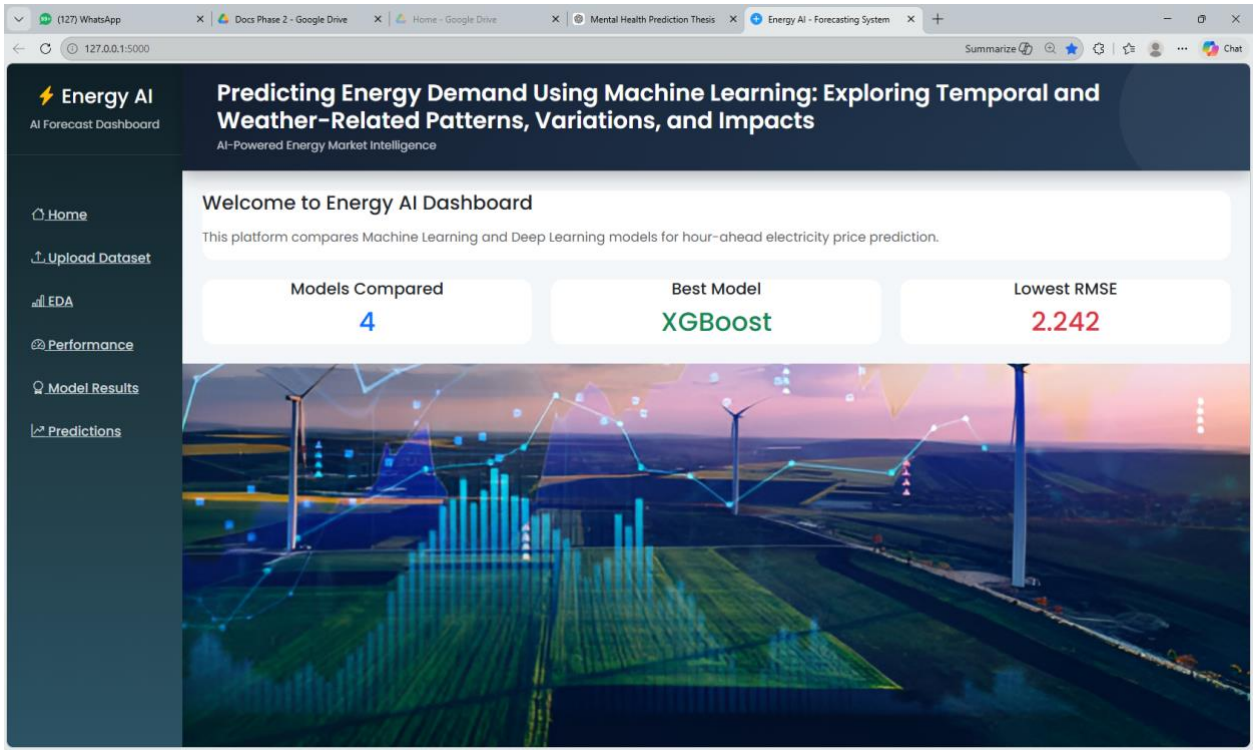
```
plt.figure(figsize=(12,6))
plt.plot(y_test_inv[:200], label="Actual Price")
plt.plot(multivariate_stacked_lstm_forecast[:200],
label="Predicted Price")
plt.title("Stacked LSTM - Hour Ahead Electricity Price
Forecast")
plt.xlabel("Time Steps")
plt.ylabel("Electricity Price")
plt.legend()
plt.grid(True)

plt.savefig("../static/performance/stacked_lstm_predicted_vs_a
ctual.png")
plt.show()

print("Stacked LSTM forecast plot saved successfully.")


# ============================================================
# 3 Calculate R2 Score
# ============================================================

r2_mult_stacked_lstm = r2_score(
    y_test_inv,
    multivariate_stacked_lstm_forecast
)

print('R2 Score of hour-ahead electricity price Stacked LSTM
forecast: {}'
      .format(round(r2_mult_stacked_lstm, 4)))


# ============================================================
# 4 Append Results to Existing RMSE DataFrame
# ============================================================

stacked_lstm_results = pd.DataFrame({
    "Model": ["Stacked LSTM"],
    "Forecast_Type": ["Hour Ahead Electricity Price"],
    "RMSE": [round(rmse_mult_stacked_lstm, 3)],
    "R2_Score": [round(r2_mult_stacked_lstm, 4)]
})

# Append to previous dataframe (XGBoost + LSTM already inside)
rmse_df = pd.concat([rmse_df, stacked_lstm_results],
ignore_index=True)

tf.keras.backend.clear_session()

encoder_decoder = tf.keras.models.Sequential([
```

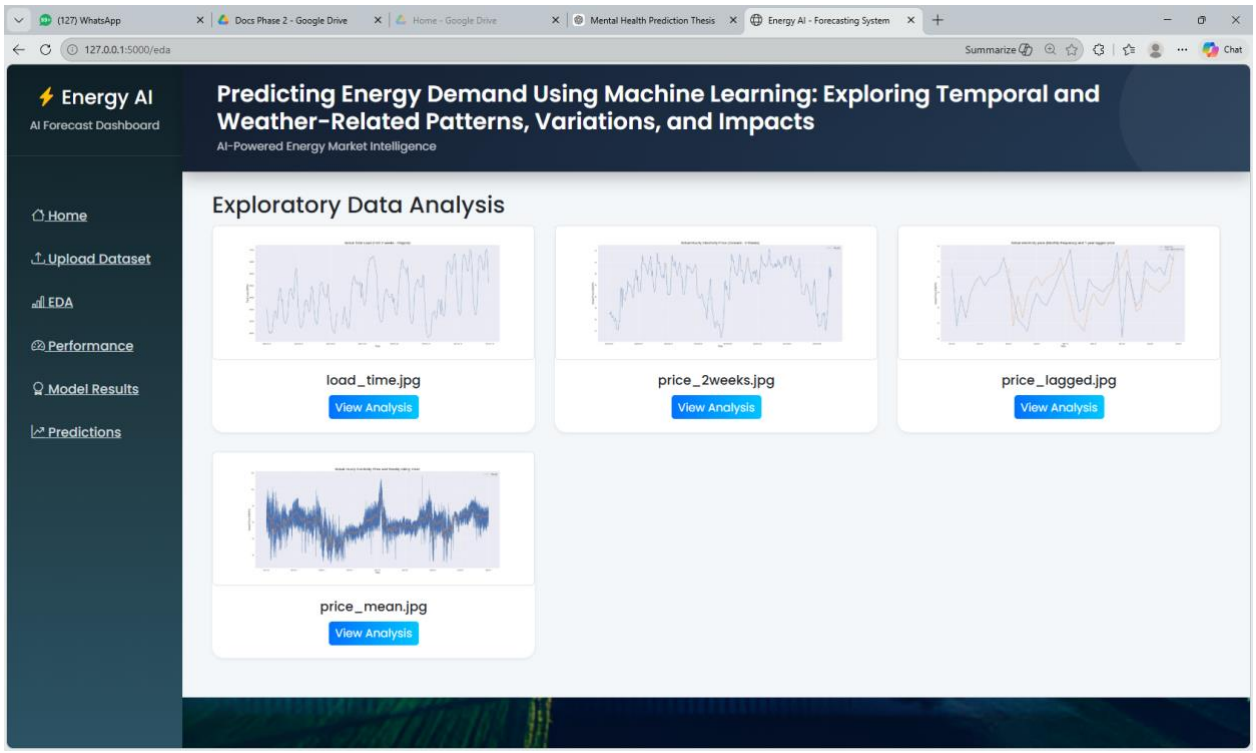
```
    LSTM(50, activation='relu', input_shape=input_shape),
    RepeatVector(past_history),
    LSTM(50, activation='relu', return_sequences=True),
    TimeDistributed(Dense(50, activation='relu')),
    Flatten(),
    Dense(25, activation='relu'),
    Dense(1)
])



model_checkpoint = tf.keras.callbacks.ModelCheckpoint(
                    'encoder_decoder.h5', save_best_only=True)

optimizer = tf.keras.optimizers.Adam(lr=1e-3, amsgrad=True)

encoder_decoder.compile(loss=loss,
                     optimizer=optimizer,
                     metrics=metric)

history = encoder_decoder.fit(train, epochs=50,
                              validation_data=validation,
                              callbacks=[early_stopping,
                                         model_checkpoint])

plot_model_rmse_and_loss(history,"Encoder-Decoder")

encoder_decoder =
tf.keras.models.load_model('encoder_decoder.h5')

forecast = encoder_decoder.predict(X_test)
encoder_decoder_forecast =
scaler_y.inverse_transform(forecast)

rmse_encoder_decoder = sqrt(mean_squared_error(y_test_inv,

encoder_decoder_forecast))
print('RMSE of hour-ahead electricity price Encoder-Decoder
forecast: {}'
      .format(round(rmse_encoder_decoder, 3)))

# Save in modern Keras format
encoder_decoder.save("../saved_models/encoder_decoder_hour_ahe
ad_model.keras")

print("Encoder-Decoder model saved successfully.")


# ==============================================================
# 2 Plot Predicted vs Actual Forecast
# ==============================================================
```

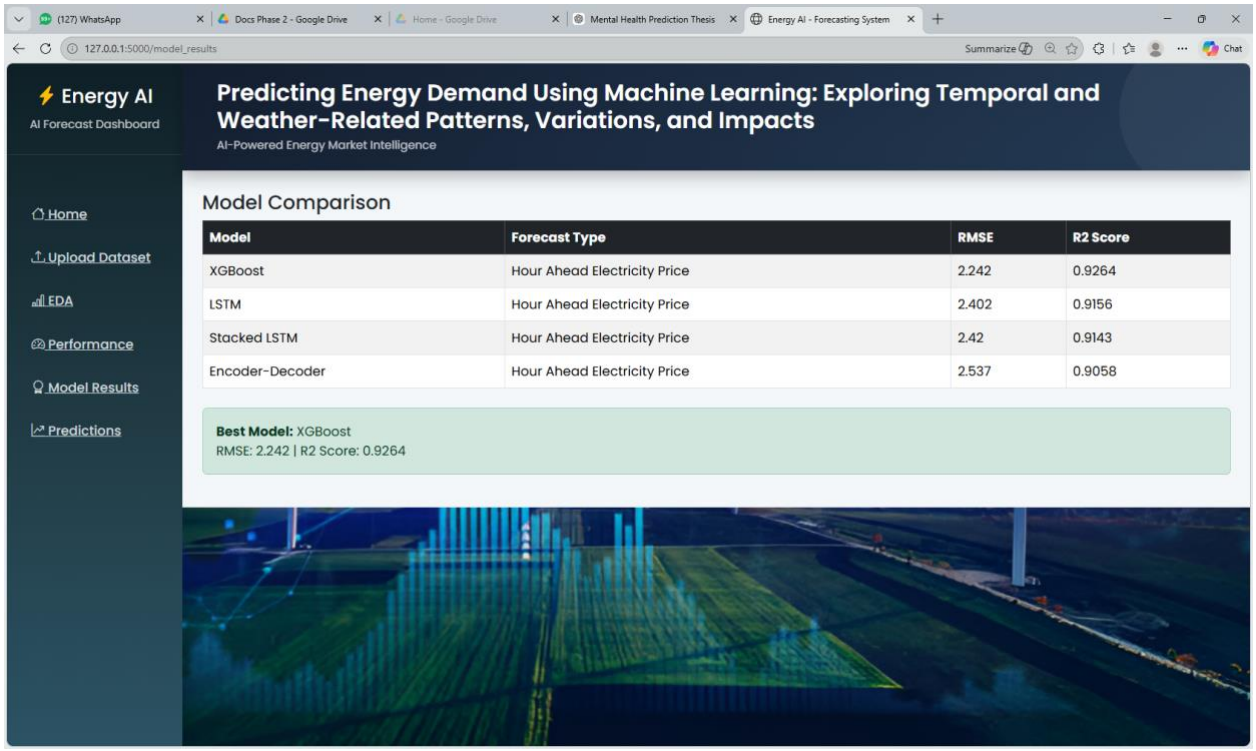
```
plt.figure(figsize=(12,6))
plt.plot(y_test_inv[:200], label="Actual Price")
plt.plot(encoder_decoder_forecast[:200], label="Predicted
Price")
plt.title("Encoder-Decoder - Hour Ahead Electricity Price
Forecast")
plt.xlabel("Time Steps")
plt.ylabel("Electricity Price")
plt.legend()
plt.grid(True)

plt.savefig("../static/performance/encoder_decoder_predicted_v
s_actual.png")
plt.show()

print("Encoder-Decoder forecast plot saved successfully.")


# ============================================================
# 3 Calculate R2 Score
# ============================================================

r2_encoder_decoder = r2_score(
    y_test_inv,
    encoder_decoder_forecast
)

print('R2 Score of hour-ahead electricity price Encoder-
Decoder forecast: {}'
      .format(round(r2_encoder_decoder, 4)))


# ============================================================
# 4 Append Results to Existing RMSE DataFrame
# ============================================================

encoder_decoder_results = pd.DataFrame({
    "Model": ["Encoder-Decoder"],
    "Forecast_Type": ["Hour Ahead Electricity Price"],
    "RMSE": [round(rmse_encoder_decoder, 3)],
    "R2_Score": [round(r2_encoder_decoder, 4)]
})

rmse_df = pd.concat([rmse_df, encoder_decoder_results],
ignore_index=True)

# Save updated comparison file
rmse_df.to_csv("../saved_models/model_rmse_results.csv",
index=False)
```

```
print("Encoder-Decoder results appended successfully.")
print(rmse_df)


# Save updated results
rmse_df.to_csv("../static/performance/model_rmse_results.csv",
index=False)

print("Stacked LSTM results appended successfully.")
print(rmse_df)
```

SCREEN SHORT



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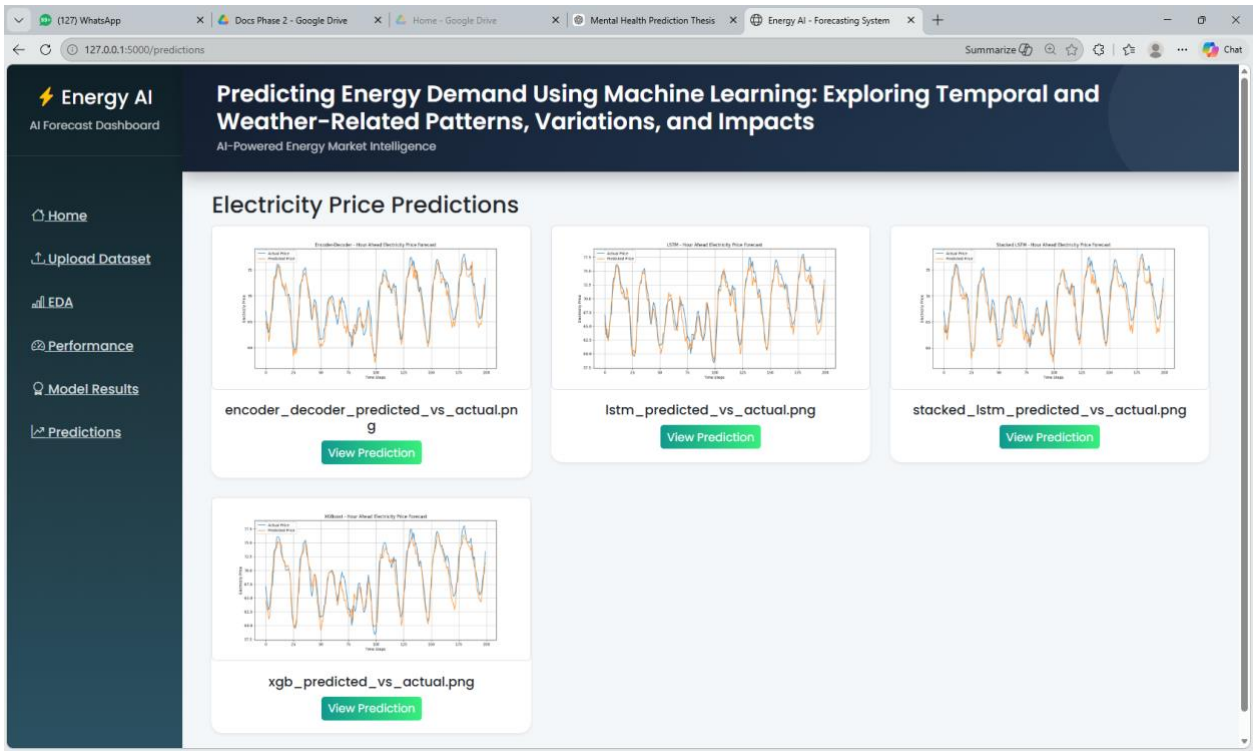
Home - Google Drive
Mental Health Prediction Thesis
Energy AI - Forecasting System
127.0.0.1:5000/model_results
Energy AI
AI Forecast Dashboard
Predicting Energy Demand Using Machine Learning: Exploring Temporal and Weather-Related Patterns, Variations, and Impacts
AI-Powered Energy Market Intelligence
Home
Upload Dataset
EDA
Performance
Model Results
Predictions
Model Comparison
Model | Forecast Type | RMSE | R2 Score
XGBoost | Hour Ahead Electricity Price | 2.242 | 0.9264
LSTM | Hour Ahead Electricity Price | 2.402 | 0.9156
Stacked LSTM | Hour Ahead Electricity Price | 2.42 | 0.9143
Encoder-Decoder | Hour Ahead Electricity Price | 2.537 | 0.9058
Best Model: XGBoost
RMSE: 2.242 | R2 Score: 0.9264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Mental Health Prediction Thesis
Energy AI - Forecasting System
127.0.0.1:5000/predictions
Energy AI
AI Forecast Dashboard
Predicting Energy Demand Using Machine Learning: Exploring Temporal and Weather-Related Patterns, Variations, and Impacts
AI-Powered Energy Market Intelligence
Home
Upload Dataset
EDA
Performance
Model Results
Predictions
Electricity Price Predictions
encoder_decoder_predicted_vs_actual.pn g
View Prediction
lstm_predicted_vs_actual.png
View Prediction
stacked_lstm_predicted_vs_actual.png
View Prediction
xgb_predicted_vs_actual.png
View Prediction

CONCLUSION

The proposed system presents an effective framework for predicting electricity prices by analyzing temporal and weather-related patterns using machine learning and deep learning models. The system integrates advanced data preprocessing techniques, feature engineering, dimensionality reduction, and time-series forecasting methods to enhance prediction accuracy. Temporal attributes such as hour, weekday, and month are extracted to capture periodic demand variations, while weather features such as temperature and temperature range provide insights into environmental influences on energy consumption and pricing. These engineered features improve the predictive capability of the forecasting models.

Multiple forecasting models including XGBoost, LSTM, Stacked LSTM, and Encoder–Decoder architectures are implemented and evaluated. The performance of each model is measured using Root Mean Squared Error (RMSE) and R^2 Score metrics. Visualization of predicted and actual electricity prices helps analyze the forecasting performance. Experimental results demonstrate that deep learning models, particularly LSTM-based architectures, are capable of capturing temporal dependencies in electricity price data. The system also provides a flexible framework that allows comparison between traditional machine learning models and deep learning approaches. Overall, the developed system successfully demonstrates the potential of machine learning techniques for analyzing complex temporal and environmental factors affecting electricity price fluctuations, thereby contributing to improved energy demand forecasting and decision-making.

Future Scope

Although the proposed system achieves promising results in electricity price forecasting, several improvements can be implemented in future research to enhance its accuracy, scalability, and applicability.

One potential improvement is the integration of advanced deep learning architectures such as Transformer models or Temporal Convolutional Networks (TCN), which have shown strong performance in time-series forecasting tasks. These models can capture long-range dependencies more effectively than traditional LSTM networks.

Another future enhancement involves incorporating additional external factors such as economic indicators, renewable energy production levels, energy market demand, and policy changes. Including these variables may significantly improve forecasting accuracy and

robustness.

Real-time energy forecasting systems can also be developed by integrating streaming data sources such as smart meters and IoT-based weather monitoring systems. This would allow continuous energy demand prediction and more responsive grid management.

Explainable artificial intelligence techniques such as SHAP or LIME could also be integrated to interpret the influence of different features on energy price predictions. This would improve transparency and help energy analysts better understand the factors affecting electricity price fluctuations.

Finally, the system could be deployed as a cloud-based or web-based application that allows energy providers, analysts, and researchers to access forecasting tools remotely. Such deployment would enable large-scale adoption of machine learning models for energy demand management and smart grid optimization.

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